# Psychological Mechanisms of the AI Placebo Effect: A Bayesian Re-Analysis of Expectation-Driven Performance in Human–Computer Interaction

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## Background & Motivation

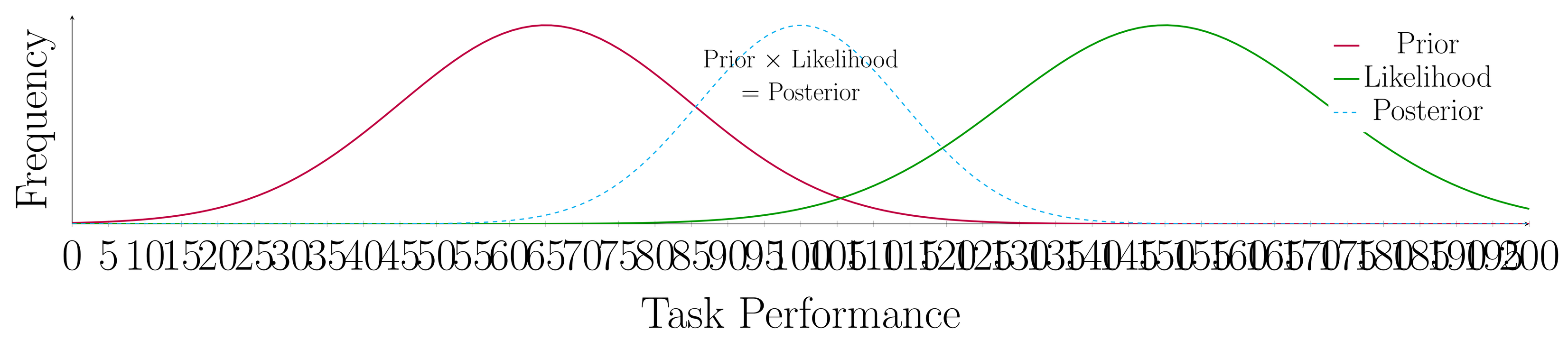
- AI systems can improve user performance not only through genuine technical capability, but also through an **"AI placebo effect"** — performance changes driven purely by users' *beliefs* about the AI, independent of what the AI actually does.
- Prior work established that this placebo effect exists behaviourally, but **why** expectations translate into objective performance change remained unexplained at a mechanistic level.
- Adopting the Bayesian brain framework of **Buchel et al. (2014)** and **Strube et al. (2023)**, we re-analysed prior placebo data by treating objective task performance as the outcome of Bayesian inference:

$$\textit{Prior Expectation} + \textit{Task Evidence} \longrightarrow \textit{Predicted Performance}$$

**Goal:** Explain the AI placebo effect in terms of the latent cognitive mechanisms underlying prior performance expectations — specifically, whether placebo shifts the *level* of prior belief, its *certainty*, or both.

## Bayesian Modelling Approach

We modelled each participant's prior expectation as a Gaussian belief that combines with task evidence (likelihood) to form a posterior predicted performance, following standard Bayesian cue-combination.




**Key latent parameters estimated:**

- `p_shift` — shift in the *mean* of prior expectation (added to stated expectancy).
- `log_var_scale` — change in the *log variance* (inverse of precision) of prior expectation.

**Three nested models** (differing in which parameters were free):

- **Shift Model** — prior mean free to shift only.
- **Precision Model** — prior precision free to change only.
- **Combined Model** — both parameters free.

## Model Fitting & Comparison

- **Parameter estimation:** Markov Chain Monte Carlo (MCMC) sampling approximated the joint posterior over all model parameters for each of the three models.
- **Implementation:** `PyMC` (`pm.sample`) with its default No-U-Turn Sampler (NUTS).
- **Sampling settings:** 4 independent chains × 2500 posterior samples, following 2500 tuning steps each; `target_accept` = 0.98; fixed random seed for reproducibility. Log-likelihoods were stored for model comparison.
- **Model comparison:** Leave-One-Out cross-validation (LOO) and the Widely Applicable Information Criterion (WAIC); models ranked by LOO (lower = better out-of-sample predictive accuracy).
- **Correlation analysis:** Pearson correlation between individual-level parameter changes and individual-level objective performance differences (Placebo − Control).

## Results I: Model Comparison (H1)

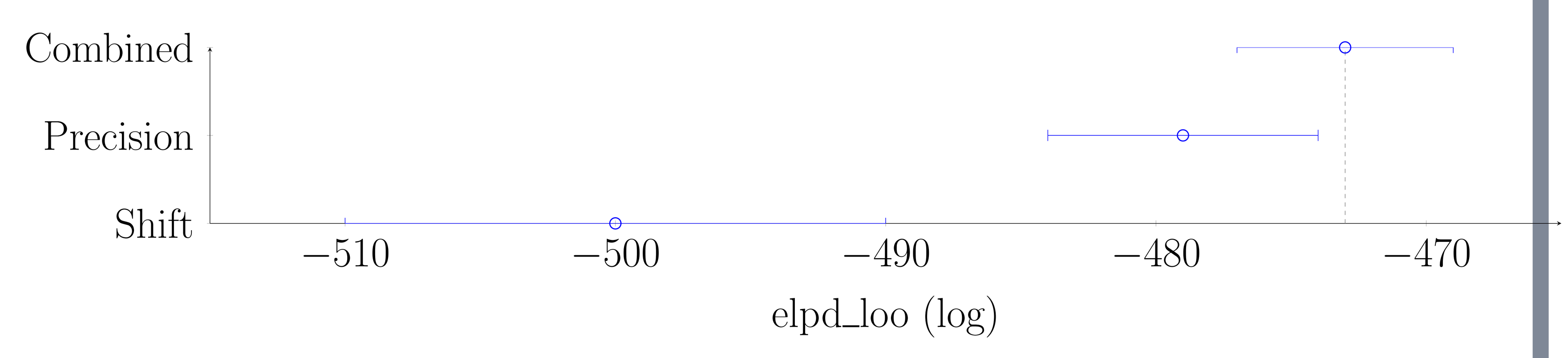



Fig. 2: Model comparison based on LOO (ELPD). Lower is worse; error bars show SE.

**Best-fitting model (H1 supported):**

- The **Combined Model** provided the best out-of-sample fit (highest / least-negative elpd_loo).
- Suggests the AI placebo effect involves modulating *both* the expected performance level *and* the certainty of that expectation, not just one or the other.

## Results II: Group-Level Parameters (H2)

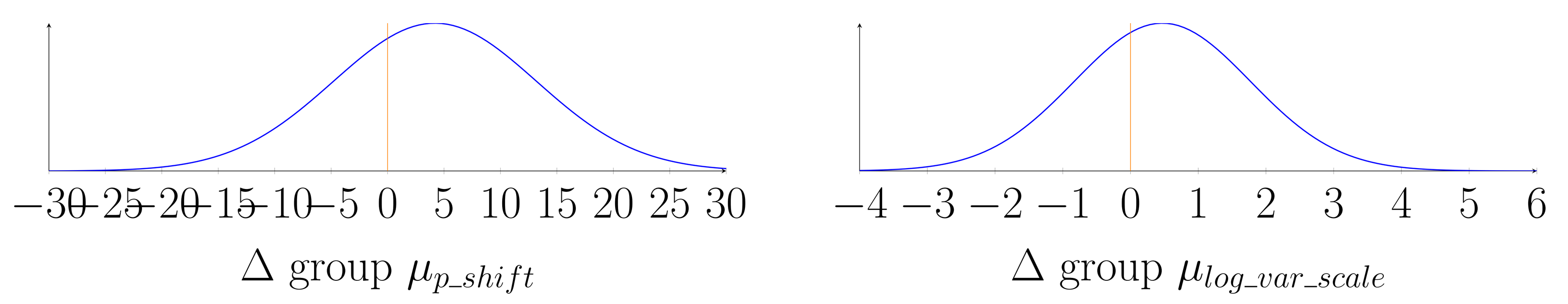



Fig. 3: Posterior distributions of group mean differences.

**H2 not supported:**

- No statistically significant *average* change in estimated prior mean shift (95% HDI: −15 to 19) or log prior variance (95% HDI: −1.7 to 3.5) at the group level.
- Mirrors the original study's finding of no group-level effect on objective performance.

## Results III: Individual-Level Correlations (H3)

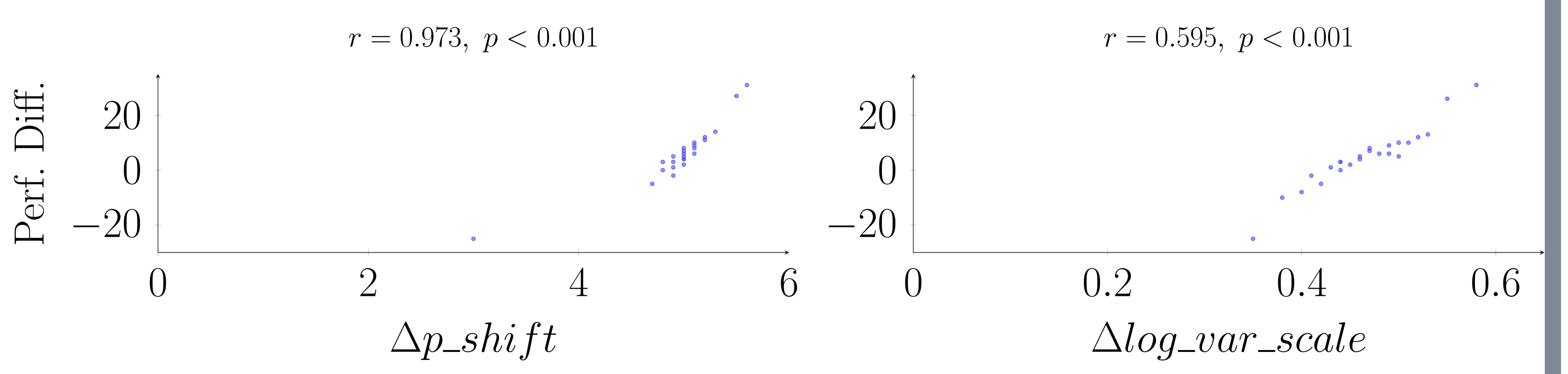



Fig. 4: Individual parameter change vs. objective performance difference (Placebo − Control).

**Prior Mean Shift (H3a, supported):** Strong positive correlation ($r = 0.973$, $p < 0.0001$) between individual `delta_p_shift` and objective performance difference — *more shift = more gain.*

**Prior Precision (H3b, opposite of prediction):** Moderate positive correlation ($r = 0.595$, $p < 0.0001$) between individual `delta_log_var_scale` and performance difference — *more variance / less precision = more gain*, an unexpected result.

## Discussion, Contributions & Future Directions

**Discussion**

- The AI placebo effect likely involves changes in both the *mean (level)* and the *precision (certainty)* of prior performance beliefs.
- There was no average group-level change in cognitive parameters; however, *individual* increases in prior mean expectation strongly predicted *individual* objective performance gains.
- Placebo's impact appears to be individual-specific and tied to prior mean shifts. Unexpectedly, *decreased* prior precision (more variance) also correlated with performance gains.

**Contributions**

- Quantified individual-specific latent cognitive parameter changes underlying the AI placebo effect.
- Explained *why* individual expectations link to performance, via prior mean shifts.
- Highlighted the importance of studying individual differences in HCI placebo effects, beyond group averages.

**Future Directions:** Investigate the counterintuitive precision effect • Replicate and generalise findings across diverse tasks, AI manipulations, and populations • Extend models to incorporate moderators (e.g., personality) and subjective ratings • Design studies with direct measures of confidence/certainty.